

MOHAMED ABDULLAH F

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Education

Aalim Muhammed Salgeh College of Engineering

2021 – 2025

Bachelor of Engineering in Computer Science and Engineering, CGPA: 7.80

Chennai, Tamil Nadu, India

Experience

Google Summer of Code - Alaska

May 2025 – September 2025

Open Source Developer

Remote

- Built the Sintra Measurement Client for RIPE Atlas, including a CLI to create ping and traceroute measurements, configuration-driven workflows, and robust result handling.
- Implemented a real-time anomaly detection Event Manager with alerts for high latency, packet loss, and latency spikes; added dry-run mode, detailed alerting, and anomaly summaries.
- Enhanced visualization by adding regional plotting support for ping measurements.

Alaskan Season of Code

May 2024 – September 2024

Remote Volunteer Intern

Remote

- I worked with the University of Alaska Anchorage Researchers under the mentorship of Pradeeban Kathiravelu, Ph.D., and David Moxley, Ph.D. for the Beehive Project.
- Designed and implemented a user profile management system using Flask with secure login and session handling.
- Developed and deployed image handling features, allowing users to dynamically upload, edit, and delete images.
- Integrated user authentication through Google OAuth 2.0 for admin access, ensuring a secure and authorized environment.
- Enhanced user experience by providing easy image management functionality.

Projects

Beehive | Python, Flask, MongoDB, Google OAuth

- Developed a data federation system to analyze behavioral health metrics.
- Improved the understanding and application of behavioral health data in healthcare practices, aiding in the enhancement of community health strategies in Alaska.
- Utilized Google OAuth 2.0 for user authentication to provide secure admin access, ensuring a protected environment.

ipscout | Golang, Cobra-CLI

- Developed IpScout, a real-time network monitoring and packet analysis tool in Go, enabling users to inspect IP traffic for both IPv4 and IPv6.
- Authored robust logging and troubleshooting mechanisms to improve usability and maintainability.
- Developed a user-friendly CLI with commands for starting monitoring, version tracking, and configuration initialization.

NetResSim | Python, RIPE Atlas

- Developed NetResSim, a Python-based simulation tool for evaluating Internet resilience under simulated regional disruptions using RIPE Atlas measurement data.
- Automated data collection, processing, and scenario analysis for resilience assessment.

Publications & Research

- Abdullah, M. F.**, Kwon, J., Bagwan, A., Gupta, I., Moxley, D., & Kathiravelu, P. “A Data Framework for Behavioral Health with Digitized Drawings and Photographs.” In *Proceedings of the 20th Annual IEEE International Systems Conference (SysCon 2026)*, April 2026. 8 pages. Accepted.

Technical Skills

Languages: Python, Golang, Java

Technologies/Frameworks: Git, Docker, Linux, GitHub, Flask, FastAPI